Environmental Product Declaration

In accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for:

Steel Plates

from

Jindal Steel & Power

Programme:	The International EPD® System, www.environdec.com
Programme operator:	EPD International AB
Licensee:	EPD India
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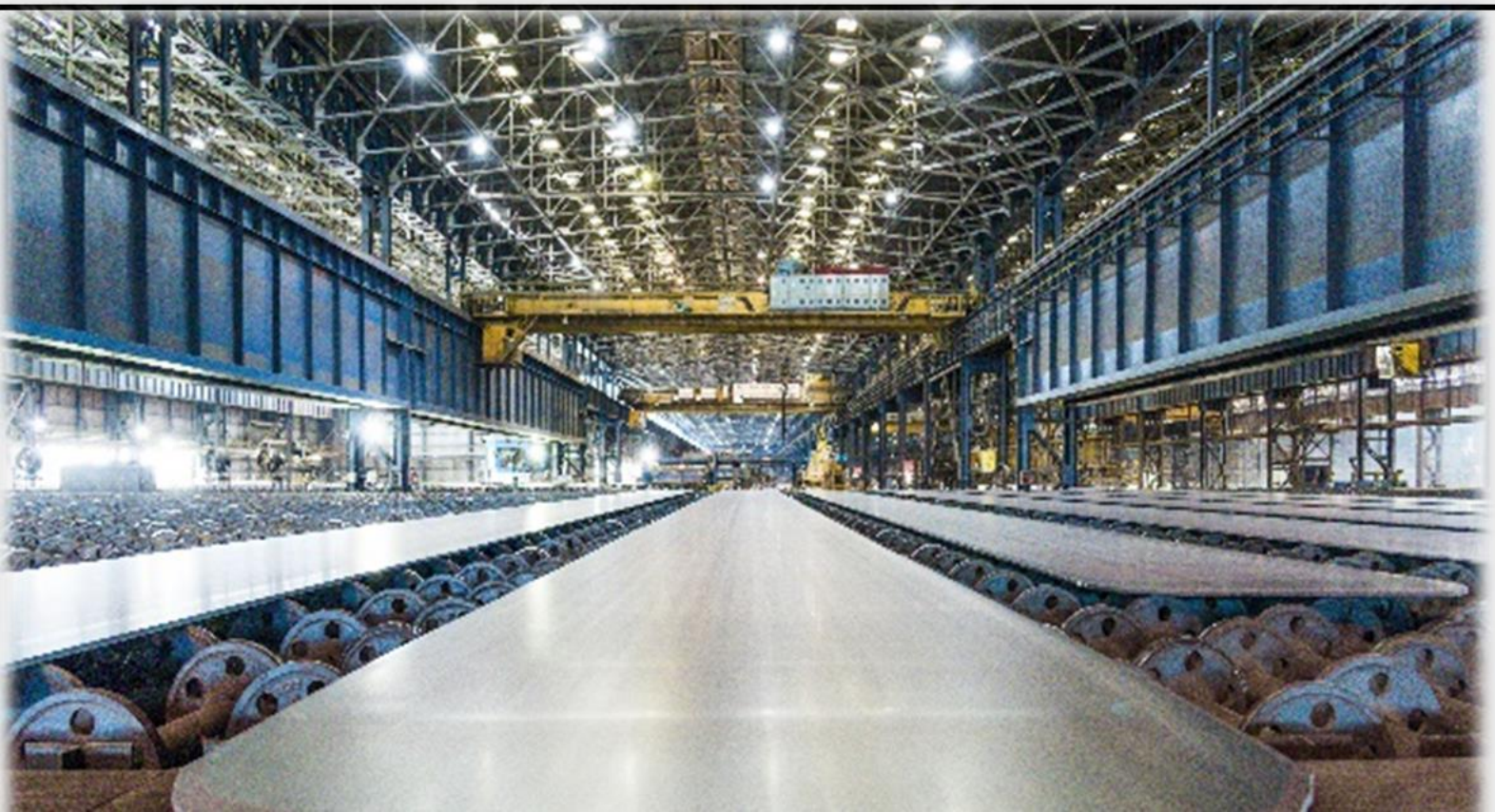
	An EPD may be updated or depublished if conditions change. To find the latest version of the EPD and to confirm its validity, see www.environdec.com.



General information

Programme information

Programme:	The International EPD® System	Indian Regional Hub
Program Operator	EPD International AB Box 210 60 SE-100 31 Stockholm Sweden	422, MIDAS, Sahar Plaza Andheri East Mumbai 400059
Website:	**www.environdec.com**	www.environdecindia.com
E-mail:	**info@environdec.com**	info@environdecindia.com

Accountabilities for PCR, LCA and independent, third-party verification
Product Category Rules (PCR)
CEN standard EN 15804 serves as the Core Product Category Rules (PCR)
Product Category Rules (PCR): *PCR 2019:14 Construction products, version 1.3.2 Published on 2023.12.08. Based on CEN standard EN 15804. CEN standard EN 15804 serve as the core PCR. UN CPC code 4121*.
PCR review was conducted by: The Technical Committee of the International EPD®System. See https://www.environdec.com/about-us/the-international-epd-system-about-the-system for a list of members.
Life Cycle Assessment (LCA)
LCA accountability: *Jindal Steel & Power.*
LCA and EPD developer: Dr. Rajesh Kumar Singh Thinkstep Sustainability Solutions Pvt. Ltd., a Sphera Company
Third-party verification
Independent third-party verification of the declaration and data, according to ISO 14025:2006, via: ☒ EPD verification by individual verifier Third party verifier: Dr. Hüdai Kara, Metsims Sustainability Consulting, 4 Clear Water Place, Oxford OX2 7NL, UK Email: hudai.kara@metsims.com Approved by: The International EPD® System
Procedure for follow-up of data during EPD validity involves third party verifier: ☐ Yes ☒ No

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but registered in different EPD programmes, or not compliant with EN 15804, may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterization factors); have equivalent content declarations; and be valid at the time of comparison.

For further information about comparability, see EN 15804 and ISO 14025.

Company information

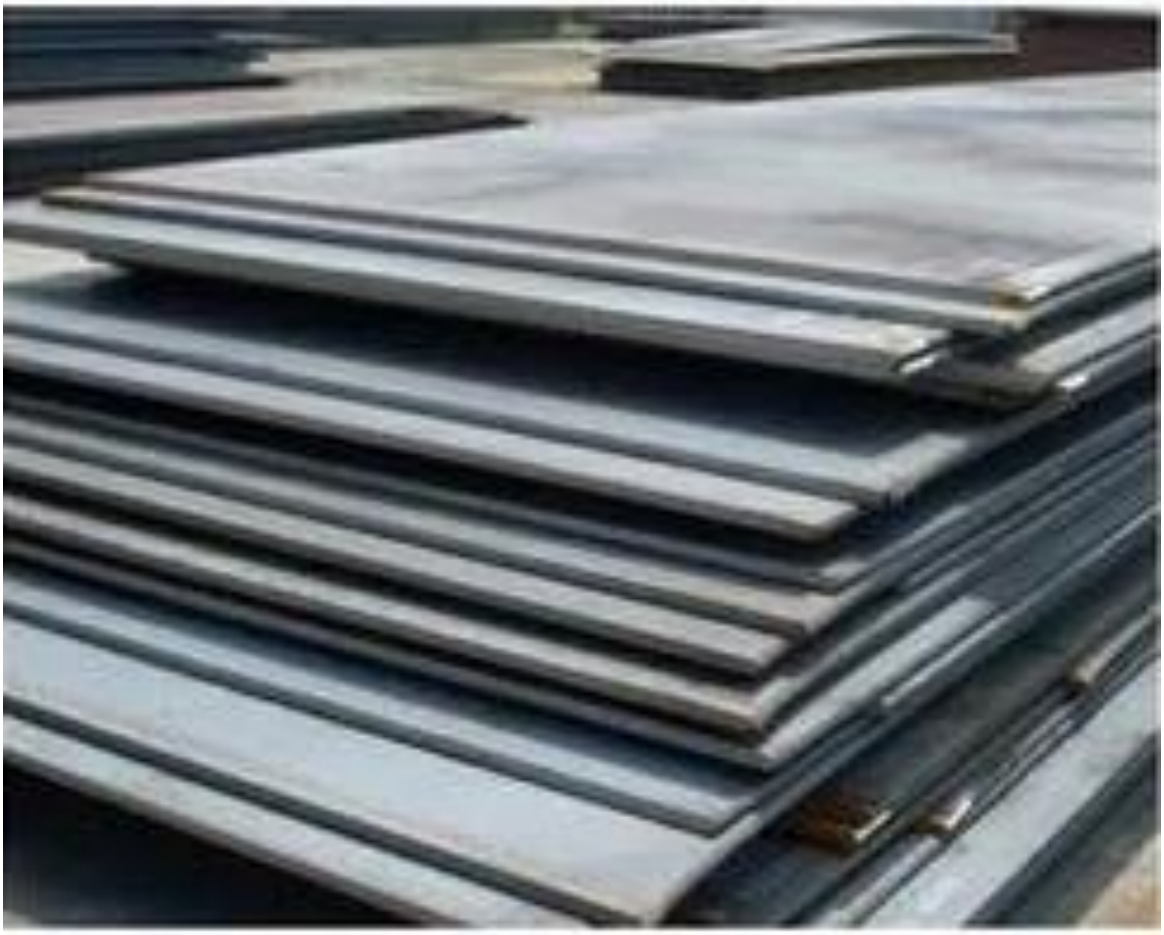
Owner of the EPD: Jindal Steel & Power.

Contact: Sanjay Nandanwar,
Email: sanjay.nandanwar@jindalsteel.com

Description of the organization:
Jindal Steel and Power is an Indian steel company which is a part of OP Jindal Group. JSP is an industrial powerhouse with a dominant presence in steel, power, mining and infrastructure sectors. The company produces best-in-class, cost-effective steel through backward and forward integration. It is undertaking ambitious expansion plans and is known as one of the most efficient & innovative steelmakers in the world.

Product-related or management system-related certifications:
All facilities at JSP are ISO 9001, ISO 14001 & ISO 45001 certified.

Name and location of production site(s):
Jindal Steel & Power
Chhendipada Road, SH-63, PO- Jindal Nagar,
Angul - 759111, Odisha, India



Product information

Product name: Steel Plates

Product identification:

Jindal Steel & Power (JSP) has left an indelible mark on Odisha's industrial landscape with its diversified operations comprising a steel plant, pellet plant, power plant, and mines. JSP has a 6 MTPA integrated steel plant comprising India's largest 4.25 MTPA Blast Furnace at Angul, Odisha. It has a 1.2 MTPA plate mill, 1.4 MTPA bar mill and an 810 MW captive power plant at Angul. Its plate mill is capable of producing 5-meter-wide plates.

UN CPC code: 4121.

Geographical scope: Global.

Product description:

Jindal Steel & Power (JSP) has the first plate mill in India, which manufactures coils and plates in 5- and 3-meter widths, respectively. Due to the cleanness of the raw materials, the soundness of the steel refining facilities, and the effectiveness of the rolling mill, these plates and coils exhibit excellent dimensional adherence and internal soundness. JSP India heavy plates are rolled in thicknesses ranging between 8 mm and 150 mm, with a maximum width of 4,900 mm and a maximum length of 24,000 mm. The maximum plate weight is 30 MT.

Manufacturing process:

The plates are manufactured through continuous casting and subsequent hot rolling of slabs. JSP is equipped with India's first state-of-the-art plate mill that produces plates. These plates are of premium quality in terms of dimensional adherence and internal soundness owing to its clean raw material, sound steel refining facilities and an efficient rolling mill. The process route is DRI-EAF-LRF-CCM-PM.

Applications:

JSP Steel Plate has wide application across verticals due to its ultra-wide range and grades produced using state of the art technology in Angul plant. The facility develops special-grade and nuclear application plates for critical applications like boilers, petroleum pipes, shipbuilding, automotive, oil exploration, earthmovers, warships, armoured vehicles, and nuclear applications.

Product Range:

Size and Grades details:

Manufacturing Unit	Product	Size (mm)	Grades
Angul	Steel Plates	Thickness: 8 - 150 Width: 1500 - 4900	• Low/Medium/High Carbon Grades • DMR Grades • Rockhard Grades • Weather Resistance Grades • API Line Pipe Grades • API Plates for offshore • Shipbuilding Grades • HSLA Grades • Boiler & Pressure vessel Grades • General engineering Grades • Quenched & Tempered Quality Grade

LCA information

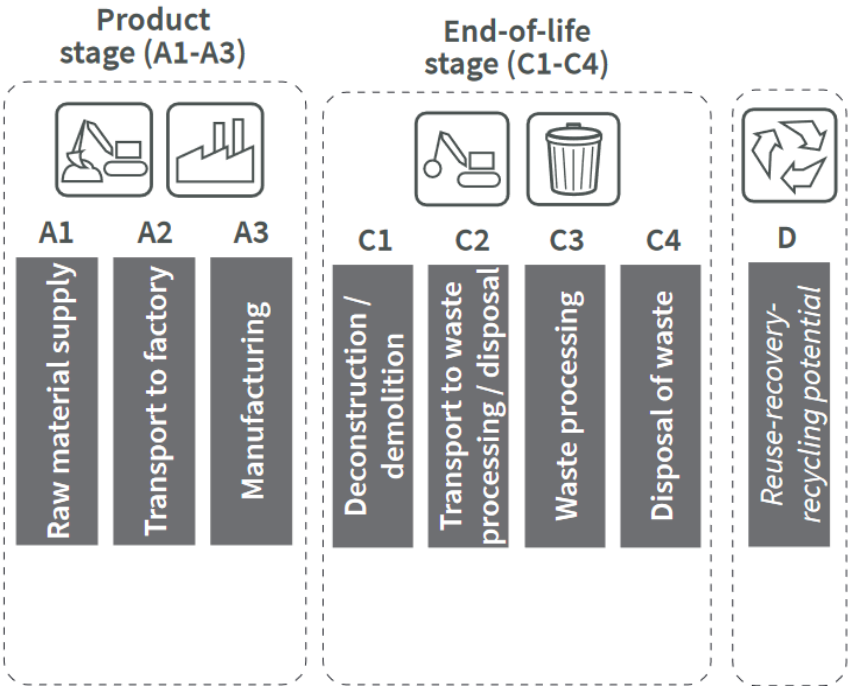
Functional unit / declared unit: 1 Tonne (1000 kg) JSP India's Steel Plates

Reference service life: Since products of that kind are not normally replaced during the service life of a building, it has been assumed that reference service life is equal to estimated Service Life of building or the structure of intended use.

Time representativeness: The collection of foreground data refers to April 2023 to March 2024.

Database(s) and LCA software used:
The background data has been taken from the latest available Sphera Managed LCA Content (MLC) 2024.1 and the LCA model was created using Sphera's LCA for Experts (LCA FE) software, version 10.7.

Description of system boundaries:
The system boundary: It includes cradle to gate stage with additional modules A1-A3, C1-C4, and module D.
Reference package used: EN 15804 reference package based on EF 3.1.



- *Module A1 to A3:*

The product stage includes provision of all materials, products and energy, as well as waste processing up to the end-of waste state or disposal of final residues during the product stage. Scrap inputs in module A1-A3 are treated as 'burden free'. In the End-of-life phase the net scrap approach has been applied towards the metal recycling benefit in module D. The standard allocation in the background dataset used for recycling benefit is based on the world steel LCI methodological report (https://worldsteel.org/wp-content/uploads/Life-cycle-inventory-methodology-report.pdf). The raw material heat mix (Hot Metal-28%, DRI- 42% and Scrap-30%) has been considered for steel plate production. The electricity in JSP India Angul facility has been modelled based on the energy sourced from a mix of Grid, Captive power plant (CPP) and the Guarantees of origin of electricity for JSP from the year 2023 – 2024. The market-based electricity modelling (Procured 698739 MWh RECs from IEX) has been done to make the DRI, SMS, Plate mill and oxygen plant renewable to achieve the low carbon impact for steel plate. The emission factor of the electricity mix for the GWP-GHG indicator is 0.13 kg CO_2eq./kWh. The packaging material is having minimal impacts, so it has not been considered in the assessment.

- *Module C1 to C4:*

Within this EPD, the modules C1-C4 are included. These modules consider the dismantling of the considered product (C1), the transportation of the dismantled components to their End of Life (EoL) destination (C2), the waste processing for recovery or recycling (C3) as well as the disposal (C4), if given. At EoL, the steel material leaves the product system in C3 for recycling in Module D. The considered EoL scenario for the steel material is 88% recycling and 12% landfill.

Category	Subcategory	Unit	Quantity
Collection process	Collected separately	kg	1000
	Collected with mixed construction waste	kg	0
Recovery	Reuse	kg	0
	Recycling	kg	880
	Landfill	kg	0
	Incineration	kg	0
	Incineration with energy recovery	kg	0
	Energy conversion efficiency rate	kg	0
Disposal	Material for final disposal	kg	120
Transport	Deconstruction site to scrap processing plant	km	0
	Scrap processing plant to site for end of waste	km	50

- *Module D:*

Module D includes any declared benefits and loads from net flows leaving the product system that have not been allocated as co-products and that have passed the end-of-waste state in the form of reuse, recovery and/or recycling potentials.

Metals are assumed to reach the end of waste status directly at the construction site. The treatment as well as net benefits and loads of reuse or recycling potentials are groupedto module D.

Potential environmental benefits are given for the net steel scrap that is produced at the end of a final product's life.
Net scrap = Amount of steel recycled at EoL – Scrap input from previous product life cycles.
The amount of steel for avoided production (0.58 t/t) is calculated by the difference between scrap recycled (0.88 t/t) and scrap input to production (0.3 t/t).

<u>Data quality assessment and declaration:</u>

Process	Source type	Source	Reference year	Data category	Share of primary data, of GWP-GHG results for A1- A3
Manufacturing of product	Collected data	EPD owner	2023	Primary data	83.68%
Generation of electricity used in manufacturing of product	Database and collected data	Sphera MLC 2023.2 and EPD owner	2022	Primary data	3.58%
Transport of raw materials to the manufacturing site	Database	Sphera MLC 2023.2	2022	Primary data	0.83%
Total share of primary data, of GWP-GHG results for A1-A3					88.09%

Note: The share of primary data is calculated based on GWP-GHG results. It is a simplified indicator for data quality that do not capture all relevant aspects of data quality. The indicator is not comparable across product categories.

Cut-off criteria:
The environmental impact of the product studied has been assessed by considering all significant processes, materials, and emissions. Excluded flows are assumed to have a negligible impact, contributing less than 5% to the cumulative impact assessment categories. The production of capital equipment, facilities, and infrastructure required for manufacture has not been considered.

Data quality and sources:
Data quality is compliant with ISO 14025:2006. All primary data were collected for the year 2024. All background data comes from the Sphera MLC 2024.1.

Allocation:
With any multi-product system, allocation rules are defined to relate the system inputs and outputs to each of the products. Several methods are documented in ISO 14040:2006 and ISO Technical Report 14049. The main co-products for Blast furnace are allocated.

Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results)

	Production			Installation		Use Stage							End-of-Life				Next product system
	Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use / application	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction / demolition	Transport to EoL	Waste processing for reuse, recovery, recycle	Disposal	Reuse, recovery, or recycling
Module	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Modules declared	X	X	X	ND	ND	ND	ND	ND	ND	ND	ND	ND	X	X	X	X	X
Geography	India												GLO	GLO	GLO	GLO	GLO
Specific data used	88.09%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation - products	0%																
Variation – Sites	0%																

X- Module declared. ND- Module not declared.

Content information

Material Properties in Steel Plate

Product Components	Weight	Post-consumer recycled material, weight-% of product	Biogenic material, weight-% of product	Biogenic material, kg C/ product or declared unit
Steel	1000 kg	20	0	0
Iron	>97%		0	0
C	0.23 % Max		0	0
Si	0.40 % Max		0	0
Mn	1.35 % Max		0	0
Cr	-		0	0
p	0.030 % Max		0	0
S	0.030 % Max		0	0
Ni	-		0	0
Cu	-		0	0
Total	**100%**		0	0

Products do not contain any substances that can be included in “Candidate List of Substances of Very High Concern for Authorization” and raw materials used are not part of the EU REACH regulation.

Results of the environmental performance indicators

The environmental performance of the functional unit of one metric tonne of JSP India Steel Plate are reported below using the parameters and units as specified in PCR 2019:14 v1.3.2.

The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.

Mandatory impact category indicators according to EN 15804+A2:2019

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-total	kg CO_2 eq.	1.65E+03	0.00E+00	5.06E+00	0.00E+00	1.81E+00	-1.01E+03
GWP-fossil	kg CO_2 eq.	1.65E+03	0.00E+00	5.06E+00	0.00E+00	1.80E+00	-1.01E+03
GWP-biogenic	kg CO_2 eq.	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
GWP-luluc	kg CO_2 eq.	4.34E-01	0.00E+00	7.57E-05	0.00E+00	1.08E-02	-1.34E-01
ODP	kg CFC -11 eq.	4.06E-08	0.00E+00	8.18E-14	0.00E+00	4.84E-12	-1.35E-09
AP	Mole of H^+ eq.	2.98E+00	0.00E+00	2.53E-02	0.00E+00	1.28E-02	-2.46E+00
EP- freshwater	kg P eq.	8.13E-04	0.00E+00	9.60E-07	0.00E+00	4.08E-06	-2.35E-04
EP- marine	kg N eq.	4.93E-01	0.00E+00	9.18E-03	0.00E+00	3.28E-03	-3.96E-01
EP- terrestrial	Mole of N eq.	5.35E+00	0.00E+00	1.01E-01	0.00E+00	3.62E-02	-3.55E+00
POCP	kg NMVOC eq.	1.42E+00	0.00E+00	2.61E-02	0.00E+00	1.00E-02	-1.61E+00
ADPE	kg Sb eq.	4.99E-04	0.00E+00	3.40E-08	0.00E+00	1.16E-07	-5.70E-03
ADPF	MJ	2.65E+04	0.00E+00	7.03E+01	0.00E+00	2.37E+01	-1.00E+04
WDP	m^3 world equiv.	2.06E+01	0.00E+00	5.88E-03	0.00E+00	2.06E-01	-6.80E+01
Acronyms	**Caption:** GWP - total = global warming potential; GWP - fossil = global warming potential (fossil fuel only); GWP - biogenic = global warming potential (biogenic); GWP - luluc = global warming potential (land use only); ODP = ozone depletion; AP = acidification terrestrial and freshwater; EP freshwater = eutrophication potential (freshwater); EP - marine = eutrophication potential (marine); EP- terrestric = eutrophication potential (terrestrial); POCP = photochemical ozone formation; ADPE = abiotic depletion potential (element); ADPF = abiotic depletion potential (fossil); WDP = water scarcity.						

Additional mandatory and voluntary impact category indicators

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-GHG*	kg CO_2 eq.	1.65E+03	0.00E+00	5.04E+00	0.00E+00	1.79E+00	-1.00E+03
Acronyms	GWP-GHG*= global warming potential (greenhouse gases) This indicator includes all greenhouse gases included in GWP-total but excludes biogenic carbon dioxide emissions and uptake and biogenic carbon stored in the product with characterization factors (CFs) based on IPCC (2013)						

Resource use indicators according to EN 15804+A2:2019

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
PERE	MJ	1.05E+04	0.00E+00	7.08E-02	0.00E+00	4.13E+00	-4.20E+03
PERM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	1.05E+04	0.00E+00	7.08E-02	0.00E+00	4.13E+00	-4.20E+03
PENRE	MJ	1.61E+03	0.00E+00	7.03E+01	0.00E+00	2.37E+01	-1.45E+02
PENRM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT	MJ	1.61E+03	0.00E+00	7.03E+01	0.00E+00	2.37E+01	-1.45E+02
SM	kg	2.00E+02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m^3	1.19E+00	0.00E+00	1.86E-04	0.00E+00	6.28E-03	-3.92E+00
Acronyms	**Caption:** PERE = Use of renewable primary energy excluding the renewable primary energy resource used as raw materials; PERM = Use of renewable primary energy as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding the non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy used as raw materials; PENRT = Total use of non-renewable primary energy resources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water						

Output flows & Waste Indicators according to EN 15804+A2:2019

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
HWD	kg	3.02E-06	0.00E+00	1.74E-10	0.00E+00	5.90E-09	-7.49E-05
NHWD	kg	1.63E+01	0.00E+00	1.00E-03	0.00E+00	1.20E+02	-1.21E+02
RWD	kg	4.25E-02	0.00E+00	7.95E-06	0.00E+00	2.48E-04	-1.10E-03
MFR	kg	0.00E+00	0.00E+00	0.00E+00	8.80E+02	0.00E+00	0.00E+00
MER	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EEE	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EET	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Acronyms	**Caption:** HWD = Hazardous waste disposed; NHWD = Non-hazardous waste disposed; RWD = Radioactive waste disposed; MFR = Materials for recycling; MER = Materials for energy recovery; EEE = Exported electrical energy; EET = Exported thermal energy						

Additional environmental performance indicators according to EN 15804+A2:2019

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
PM	Disease incidences	4.19E-05	0.00E+00	5.57E-07	0.00E+00	1.60E-07	-2.31E-05
IR	kBq U235 eq.	3.55E+00	0.00E+00	7.22E-04	0.00E+00	2.87E-02	-2.26E+01
ETF	CTU_e	1.78E+03	0.00E+00	2.96E+01	0.00E+00	1.58E+01	-5.26E+02
HTP-c	CTU_h	4.53E-08	0.00E+00	4.80E-10	0.00E+00	3.22E-10	-4.12E-07
HTP-nc	CTU_h	1.22E-06	0.00E+00	1.02E-08	0.00E+00	1.25E-08	-1.96E-06
SQP	Pt	5.10E+02	0.00E+00	7.19E-02	0.00E+00	6.52E+00	-1.30E+02
Acronyms	**Caption:** PM = Particulate matter emissions; IR = Ionising radiation, human health; ETF= Eco-toxicity (freshwater); HTP-c = Human toxicity, cancer effects; HTP-nc = Human toxicity, non-cancer effects; SQP = Soil quality potential/Land use related impacts						

References

- General Programme Instructions of the International EPD® System. Version 4.0.
- PCR 2019:14. Construction Products, Version 1.3.2
- Sustainability of construction works - Environmental product declarations - Methodology for selection and use of generic data; CEN/TR 15941:2010
- EN 15804: EN 15804:2012+A2:2019: Sustainability of construction works -Environmental Product Declarations - Core rules for the product category of construction products.
- EN ISO 14025: EN ISO 14025:2011-10 Environmental labels and declarations - Type III environmental declarations - Principles and procedures
- EN ISO 14040: EN ISO 14040:2009-11 Environmental management - Life cycle assessment - Principles and framework
- EN ISO 14044: EN ISO 14044:2006-10 Environmental management - Life cycle assessment - Requirements and guidelines.
- LCA FE: LCA FE Software System and Database for Life Cycle Engineering, Sphera Solution GmbH, Leinfelden-Echterdingen, 2024 (https://sphera.com/solutions/product-stewardship/life-cycle-assessment-software-and-data/managed-lca-content/)
- WSI and Eurofer's Co-Product Allocation Methodology 2014 – A methodology to determine the LCI of Steel Industry Co-products.
- World Steel Association- CO_2 Data Collection User Guide, Version 9 (May 2019).